

Introdução ao Wireshark

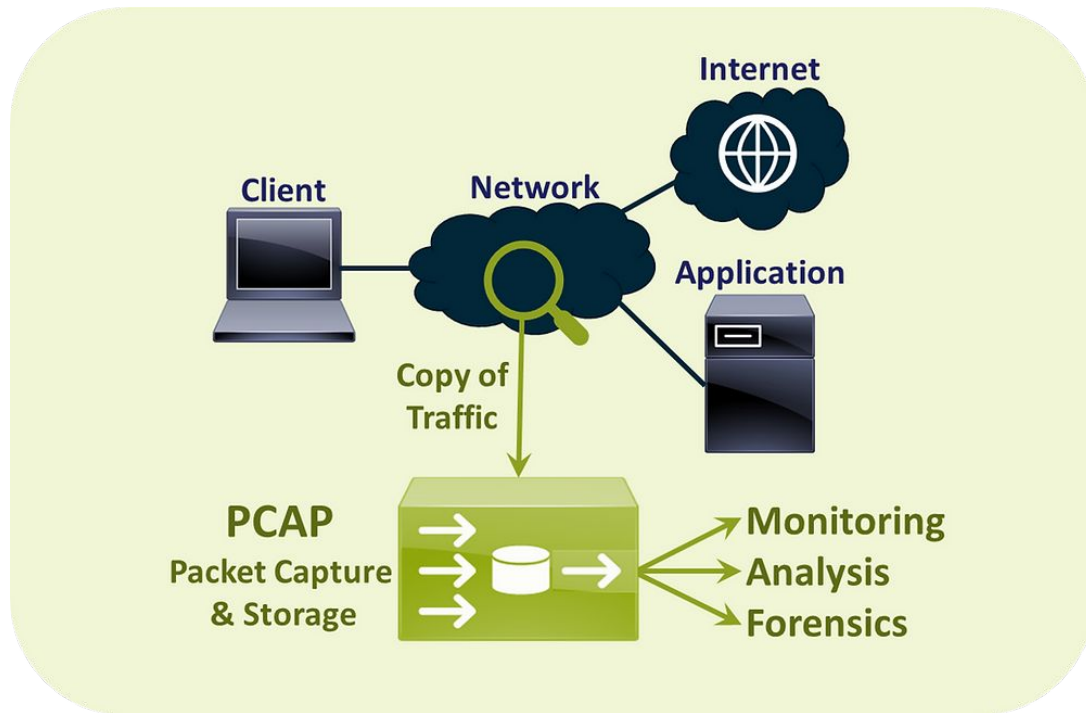
Prof. Kelvin Lopes Dias
Estagiária Maria Katarine S. Barbosa

O que é o Wireshark?

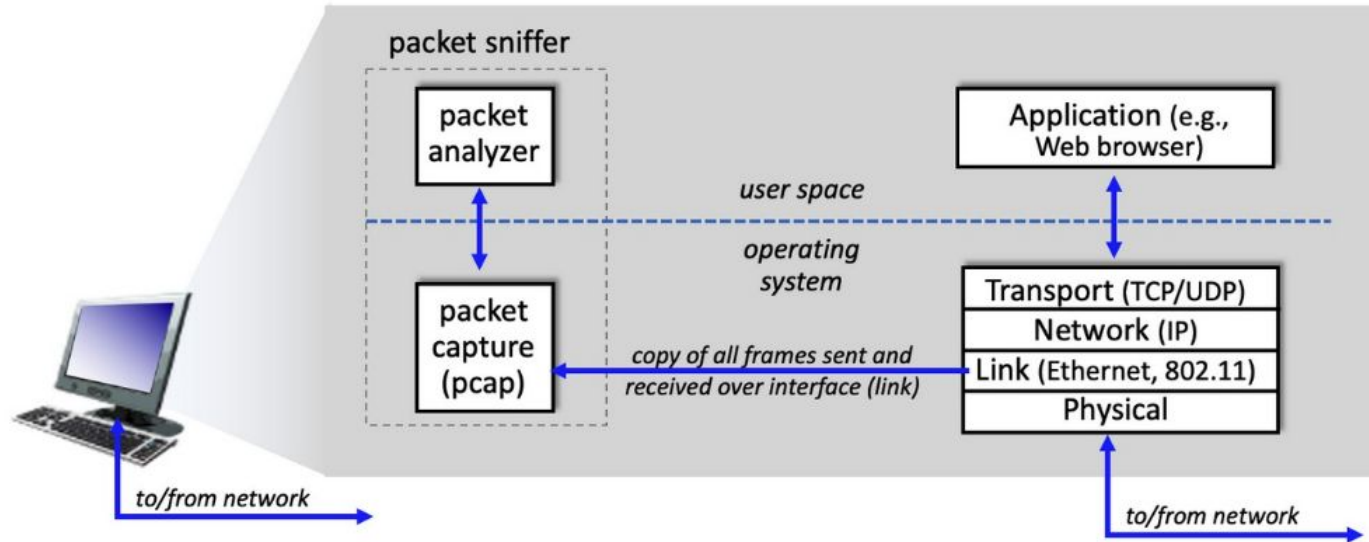


Packet Sniffer

- Identificar problemas de rede;
- Prevenir ataques;
- Encontrar programas mal intencionados;
- Analisar os possíveis gargalos de rede;
- Entender/Visualizar o funcionamento da rede.

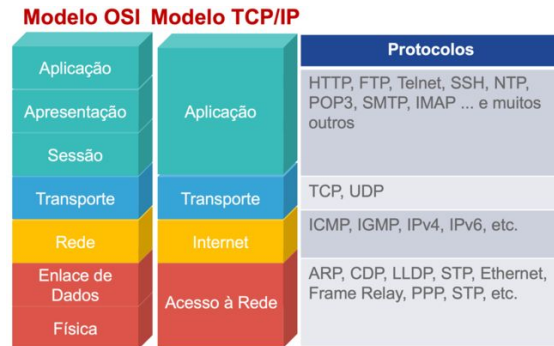


Estrutura de um Packet Sniffer



Visualização dos protocolos

1. Camada Física (Layer 1):
 - a. Principais Campos e Funções: Bits, sinais elétricos, especificações de hardware.
 - b. Serviço para a Camada Superior: Transmissão de bits através do meio físico.
2. Camada de Enlace de Dados (Layer 2):
 - a. Principais Campos e Funções: Quadros (frames), endereços MAC, controle de acesso ao meio.
 - b. Serviço para a Camada Superior: Entrega de frames entre dispositivos diretamente conectados.
3. Camada de Rede (Layer 3):
 - a. Principais Campos e Funções: Pacotes, endereços IP, roteamento.
 - b. Serviço para a Camada Superior: Roteamento de pacotes entre redes diferentes.
4. Camada de Transporte (Layer 4):
 - a. Principais Campos e Funções: Segmentos, portas, controle de fluxo.
 - b. Serviço para a Camada Superior: Entrega de dados fim a fim, controle de fluxo e correção de erros.
5. Camada de Aplicação (Layer 5):
 - a. Principais Campos e Funções: Diálogos, controle de aplicação, sincronização.



Instalando o Wireshark



NEWS Get Acquainted ▾ Get Help ▾ Develop ▾

[Link Download](#)

Download Wireshark

The current stable release of Wireshark is 3.6.6. It supersedes all previous releases. You can also download the latest development release (3.7.1) and documentation.

Stable Release (3.6.6)

- Windows Installer (64-bit)
- Windows Installer (32-bit)
- Windows PortableApps® (64-bit)
- Windows PortableApps® (32-bit)
- macOS Arm 64-bit .dmg
- macOS Intel 64-bit .dmg
- Source Code

Old Stable Release (3.4.14)

Development Release (3.7.1)

Documentation

Not What You're Looking For?

Sponsors

Always-on, scalable
Packet Capture
that integrates
with all your tools



FMADIO
10G 40G 100G PACKET CAPTURE

Never Drop Packets!
100Gbps 148Mpps sustained 24/7

Line Rate Full Packet Capture
Hardware System

sysdig
Creator of Ftrace

Sysdig Welcomes
Gerald and the
Wireshark Community

LEARN MORE

SCS

WIRESHARK
Authorized Training Partner

Official TCP/IP Troubleshooting Course
Training & Wireshark Tools

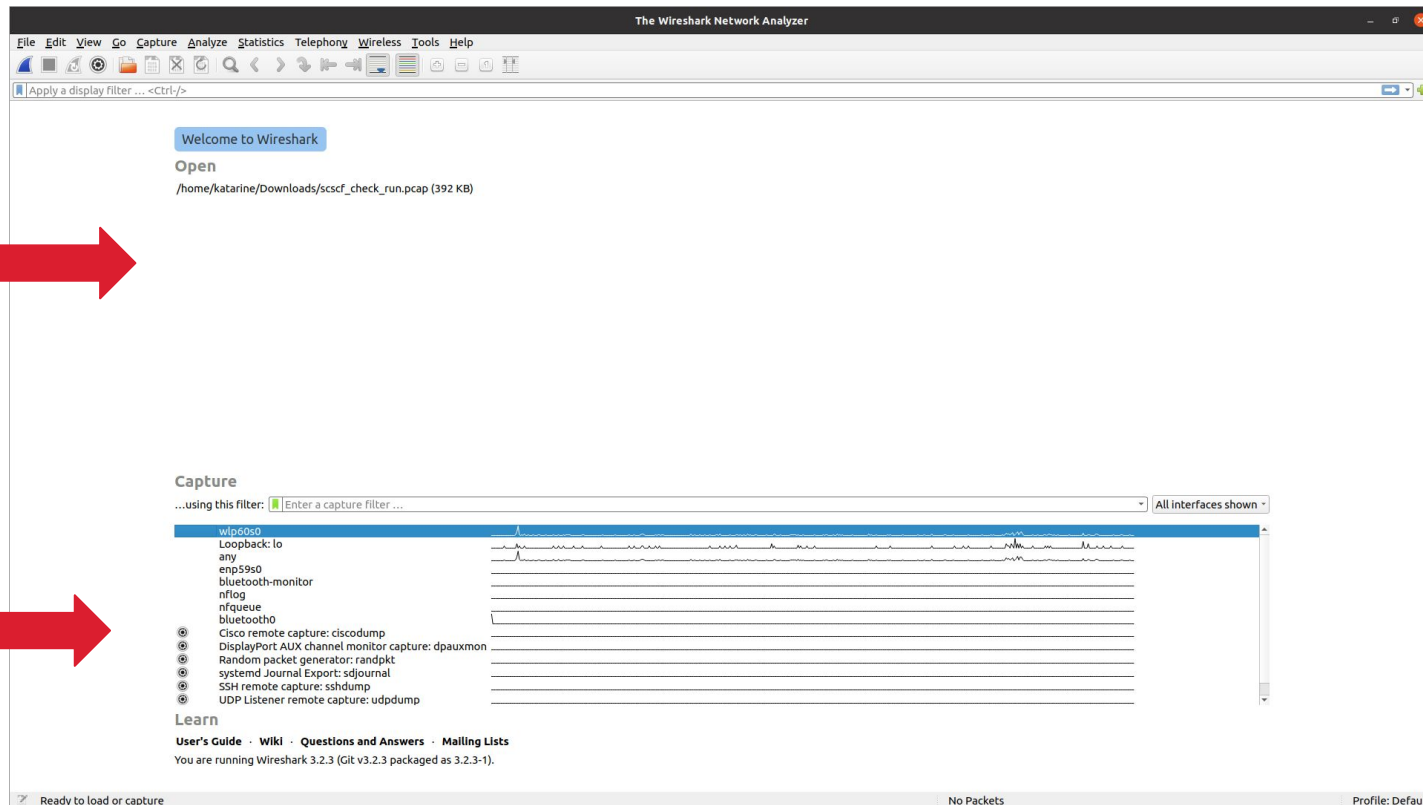
www.scs.training

Visualização inicial

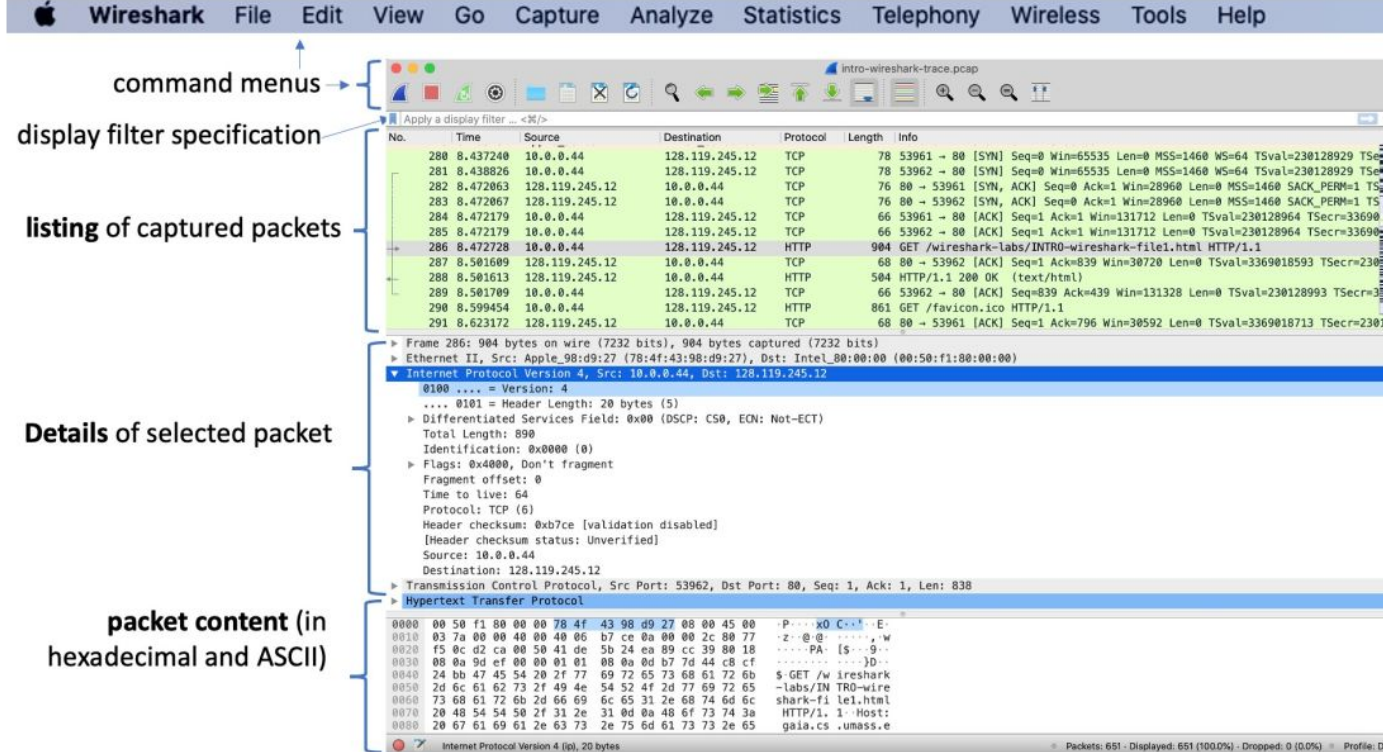
Arquivos abertos recentemente



Lista de interfaces para a realização da captura de pacotes



Capturando pacotes



command menus →

display filter specification →

listing of captured packets →

Details of selected packet →

packet content (in hexadecimal and ASCII) →

Wireshark interface showing packet capture and details of a selected packet.

The interface displays a list of captured packets with columns: No., Time, Source, Destination, Protocol, Length, and Info.

The selected packet (No. 286) is an HTTP GET request to /wireshark-labs/INTRO-wireshark-file1.html. The details pane shows the packet structure:

- Frame 286: 904 bytes on wire (7232 bits), 904 bytes captured (7232 bits)
- Ethernet II, Src: Apple_98:d9:27 (78:4f:43:98:d9:27), Dst: Intel_80:00:00 (00:50:f1:80:00:00)
- Internet Protocol Version 4, Src: 10.0.0.44, Dst: 128.119.245.12
 - Version: 4
 - Header Length: 20 bytes (5)
 - Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 - Total Length: 890
 - Identification: 0x0000 (0)
 - Flags: 0x4000, Don't fragment
 - Fragment offset: 0
 - Time to live: 64
 - Protocol: TCP (6)
 - Header checksum: 0xb7ce [validation disabled]
 - [Header checksum status: Unverified]
 - Source: 10.0.0.44
 - Destination: 128.119.245.12
- Transmission Control Protocol, Src Port: 53962, Dst Port: 80, Seq: 1, Ack: 1, Len: 838
- Hypertext Transfer Protocol

The packet content (hexadecimal and ASCII) is displayed at the bottom:

```

0000  00 50 f1 80 00 00 78 4f 43 98 d9 27 00 00 45 00  .P...x0.C...E.
0010  03 7a 00 00 40 00 40 06 b7 ce 0a 00 00 2c 80 77  .Z..@.....W
0020  f5 0c d2 ca 00 50 41 de 5b 24 ea 89 cc 39 80 18  ....PA: [$..9..
0030  08 0a 9d ef 00 00 01 01 08 0a 0d b7 7d 44 c8 cf  ....D.....
0040  24 bb 47 45 54 20 2f 77 69 72 65 73 68 61 72 6b  $GET /w ireshark
0050  2d 6c 61 62 73 2f 49 4e 54 52 4f 20 77 69 72 65  -labs/IN TRO-wire
0060  73 68 61 72 6b 2d 66 69 6c 65 31 2e 68 74 6d 6c  shark-fi le1.html
0070  20 48 54 54 50 2f 31 2e 31 0d 0a 48 6f 73 74 3a  HTTP/1.1 Host:
0080  20 67 61 69 61 2e 63 73 2e 75 6d 61 73 73 2e 65  gaia.cs .umass.e
  
```

Visão Geral

1. Frame:

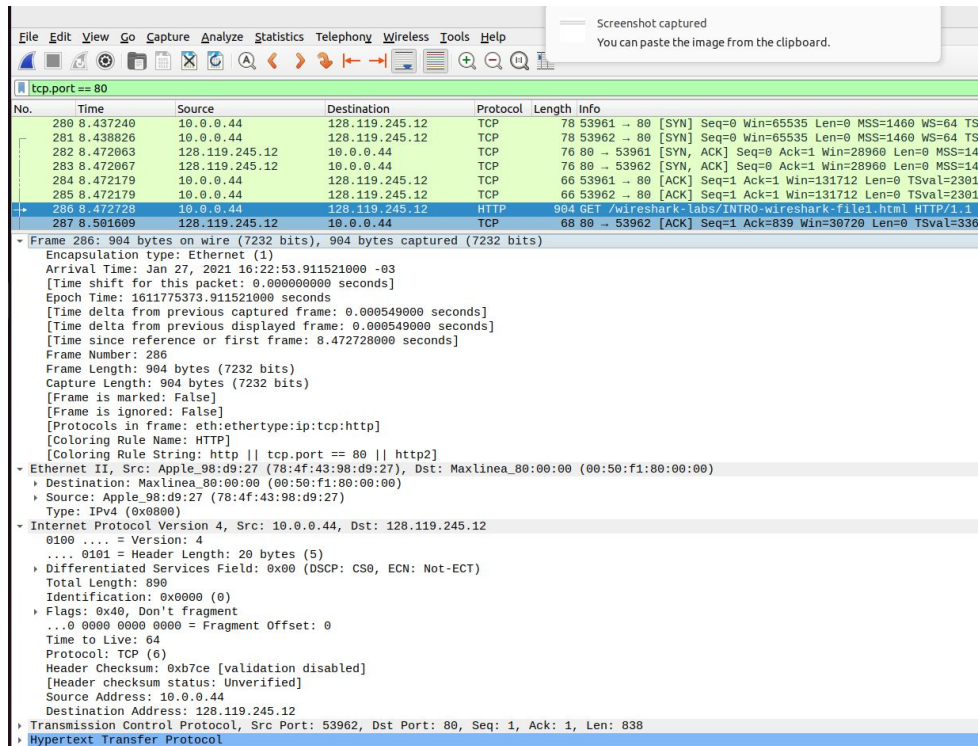
- Frame Number: Número sequencial do quadro.
- Frame Length: Tamanho do quadro em bytes.
- Encapsulation Type: Tipo de quadro (Ethernet, Wi-Fi, etc.).

2. Ethernet 2:

- Destination MAC Address: Endereço MAC de destino.
- Source MAC Address: Endereço MAC de origem.
- Type: Tipo de protocolo encapsulado (IPv4, ARP, etc.).

3. IPv4:

- Version: Versão do protocolo IP (IPv4 ou IPv6).
- Header Length: Tamanho do cabeçalho IP.
- Source/Destination IP: Endereços IP de origem e destino.
- Protocol: Protocolo encapsulado (TCP, UDP, ICMP, etc.).
- Time to Live (TTL): Tempo que um pacote tem permissão para permanecer na rede antes de ser descartado. É decrementado a cada roteador que o pacote atravessa.



Screenshot captured
You can paste the image from the clipboard.

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.port == 80

No.	Time	Source	Destination	Protocol	Length	Info
280	8.437240	10.0.0.44	128.119.245.12	TCP	78	53961 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TS=
281	8.438826	10.0.0.44	128.119.245.12	TCP	78	53962 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TS=
282	8.472063	128.119.245.12	10.0.0.44	TCP	76	80 → 53961 [SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=14
283	8.472067	128.119.245.12	10.0.0.44	TCP	76	80 → 53962 [SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=14
284	8.472179	10.0.0.44	128.119.245.12	TCP	66	53961 → 80 [ACK] Seq=1 Ack=1 Win=131712 Len=0 TSval=2301
285	8.472179	10.0.0.44	128.119.245.12	TCP	66	53962 → 80 [ACK] Seq=1 Ack=1 Win=131712 Len=0 TSval=2301
286	8.472728	10.0.0.44	128.119.245.12	HTTP	904	GET /wireshark-labs/INTRO-wireshark-file1.html HTTP/1.1
287	8.501609	128.119.245.12	10.0.0.44	TCP	68	80 → 53962 [ACK] Seq=1 Ack=639 Win=30720 Len=0 TSval=330

Frame 286: 904 bytes on wire (7232 bits), 904 bytes captured (7232 bits)

Encapsulation type: Ethernet II (1)

Arrival Time: Jan 27, 2021 16:22:53.911521000 -03

[Time shift for this packet: 0.000000000 seconds]

Epoch Time: 1611775373.911521000 seconds

[Time delta from previous captured frame: 0.000549000 seconds]

[Time delta from previous displayed frame: 0.000549000 seconds]

[Time since reference or first frame: 8.472728000 seconds]

Frame Number: 286

Frame Length: 904 bytes (7232 bits)

Capture Length: 904 bytes (7232 bits)

[Frame is marked: False]

[Frame is ignored: False]

[Protocols in frame: eth:ethertype:ip:tcp:http]

[Coloring Rule String: http || tcp.port == 80 || http2]

Ethernet II, Src: Apple_98:d9:27 (78:4f:43:98:d9:27), Dst: Maxlinea_80:00:00 (00:50:f1:80:00:00)

Destination: Maxlinea_80:00:00 (00:50:f1:80:00:00)

Source: Apple_98:d9:27 (78:4f:43:98:d9:27)

Type: IPv4 (0x0800)

Internet Protocol Version 4, Src: 10.0.0.44, Dst: 128.119.245.12

0100 ... = Version: 4

... 0101 = Header Length: 20 bytes (5)

Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 890

Identification: 0x0000 (0)

Flags: 0x40, Don't fragment

... 0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 64

Protocol: TCP (6)

Header checksum: 0xb7ce [validation disabled]

[Header checksum status: Unverified]

Source Address: 10.0.0.44

Destination Address: 128.119.245.12

Transmission Control Protocol, Src Port: 53962, Dst Port: 80, Seq: 1, Ack: 1, Len: 838

Hypertext Transfer Protocol

Visão Geral

4. TCP:

- Source/Destination Port: Portas de origem e destino.
- Sequence/Acknowledgment Numbers: Números de sequência e de reconhecimento.
- Flags: Indicadores de controle (SYN, ACK, FIN, etc.).
- Window Size: Tamanho da janela de recepção.

5. HTTP:

- Hypertext Transfer Protocol: Informações específicas do protocolo HTTP.
- Method: Método da requisição (GET, POST, etc.).
- Status Code: Código de status da resposta.
- Host/URI: Host e URI da requisição.
- Data: Conteúdo da mensagem HTTP.

tcp.port == 80						
No.	Time	Source	Destination	Protocol	Length	Info
280	8.437240	10.0.0.44	128.119.245.12	TCP	78	53961 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TS
281	8.438826	10.0.0.44	128.119.245.12	TCP	78	53962 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TS
282	8.472863	128.119.245.12	10.0.0.44	TCP	78	80 → 53961 [SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=14
283	8.472867	128.119.245.12	10.0.0.44	TCP	78	80 → 53962 [SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=14
284	8.472179	10.0.0.44	128.119.245.12	TCP	66	53961 → 80 [ACK] Seq=1 Ack=1 Win=131712 Len=0 TSval=2381
285	8.472179	10.0.0.44	128.119.245.12	TCP	66	53962 → 80 [ACK] Seq=1 Ack=1 Win=131712 Len=0 TSval=2381
286	8.472728	10.0.0.44	128.119.245.12	HTTP	904	GET /wireshark-labs/INTRO-wireshark-file1.html HTTP/1.1
287	8.501609	128.119.245.12	10.0.0.44	TCP	68	80 → 53962 [ACK] Seq=1 Ack=839 Win=30720 Len=0 TSval=336
Frame 286: 904 bytes on wire (7232 bits), 904 bytes captured (7232 bits) Ethernet II, Src: Apple_98:d9:27 (78:4f:43:98:d9:27), Dst: Maxlinea_80:00:00 (00:50:f1:80:00:00) Internet Protocol Version 4, Src: 10.0.0.44, Dst: 128.119.245.12 Transmission Control Protocol, Src Port: 53962, Dst Port: 80, Seq: 1, Ack: 1, Len: 838 Source Port: 53962 Destination Port: 80 [Stream index: 5] [Conversation completeness: Incomplete, DATA (15)] [TCP Segment Len: 838] Sequence Number: 1 (relative sequence number) Sequence Number (raw): 1105091364 [Next Sequence Number: 839 (relative sequence number)] Acknowledgment Number: 1 (relative ack number) Acknowledgment number (raw): 3934899257 1000 = Header Length: 32 bytes (8) Flags: 0x018 (PSH, ACK) Window: 2858 [Calculated window size: 131712] [Window size scaling factor: 64] Checksum: 0x9def [unverified] [Checksum Status: Unverified] Urgent Pointer: 0 Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps [Timestamps] [SEQ/ACK analysis] TCP payload (838 bytes) Hypertext Transfer Protocol GET /wireshark-labs/INTRO-wireshark-file1.html HTTP/1.1\r\n Host: gaia.cs.umass.edu\r\n User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10.15; rv:84.0) Gecko/20100101 Firefox/84.0\r\n Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,*/*;q=0.8\r\n Accept-Language: en-US,en;q=0.5\r\n Accept-Encoding: gzip, deflate\r\n DNT: 1\r\n Connection: keep-alive\r\n [truncated]Cookie: _ga=GA1.2.539094814.1610028235; _fbp=fb.1.1581132068384.462218827; _hjid=721b807f-ada0-4c94-9043-2eb27649f9ca;						
0020	f5 0c d2 ca 00 50 41 de	5b 24 ea 89 cc 39 80 18	...PA- [\$...9...			
0030	08 0a 9d ef 00 00 01 01	08 0a 0d b7 7d 44 c8 cf	)D...			
0040	24 bb 47 45 54 20 2f 77	69 72 65 73 68 61 72 6b	\$ GET /w ireshark			
0050	2d 6c 61 62 73 2f 49 4e	54 52 4f 2d 77 69 72 65	-labs/IN TRO-wire			

Exercício

1. Iniciar o Wireshark;
2. Iniciar captura de pacotes;
3. Acessar o link disponível no pdf anexado no *classroom*;
4. Parar a captura.

Aplicando Filtros

- Filtrando protocolos:
 - Inserir o nome do protocolo (tcp, udp, arp, dhcp, icmp, etc)
- Filtrando IP:
 - `ip.addr == 127.0.0.1` (Pacotes que contenham o IP especificado);
 - `ip.src == 127.0.0.1` (Pacotes cuja origem seja o endereço IP especificado);
 - `ip.dst == 127.0.0.1` (Pacotes cujo destino seja o endereço IP especificado).
- Filtrando endereços MAC:
 - `eth.addr == "endereço MAC"`
 - `eth.src == "endereço MAC"`
 - `eth.dst == "endereço MAC"`
- [Outros Filtros](#)

Aplicando Múltiplos Filtros

- Junção (or)
 - `eth.src == "endereço MAC" or tcp`
- Interseção (and)
 - `ip.addr == 192.168.153.90 and tcp`
- Impedir que valor seja exibido
 - `!(ip.addr == 127.0.0.1) ou not (ip.addr == 127.0.0.1)` (Informações do IP especificado não serão exibidas)

Dúvidas ?



Introdução ao Wireshark

Prof. Kelvin Lopes Dias
Estagiária Maria Katarine S. Barbosa